Smart Government Service Delivery Platform

CivicFlow

Group 1
MIT 820 Software Development Project I

Final presentation 28 November 2026 Application 1.0.4

The service delivery problem

Uncertain receipt and progress

Citizens need a recorded request and a clear next step.

Disconnected information and evidence

Forms, documents and follow-up should remain within one record.

Unclear processing responsibility

A department and authorised officer must own the next action.

Private records need controlled access

Transparency for the owner must preserve other users’ confidentiality.

Aim and project objectives

Online citizen access

Registration, complaints, permit applications and private supporting evidence.

Controlled service processing

Department routing, review, information requests and reasoned decisions.

Visible progress and communication

Tracking, notifications, an external delivery outbox and optional assistance.

Reviewable implementation evidence

Granular permissions, repeatable checks, manuals and deployment material.

Group members and allocation

Member	Registration number	Sub team
Aminu Idris	2630802400002	Development
Saidu Mustapha Suleman	2630802400001	Documentation
Bolaji Watson	2630802400009	Project management
John Ojuekaiye Mayowa	2630802400008	Development
Egbue Odira Celestine	2630803400003	Documentation

Scope and default roles

Role	Record scope	Default responsibility
Citizen	Own records	Submit, reply, track and give feedback
Service officer	Assigned department	Review, decide and communicate
Department head	Assigned department	Department work and CSV oversight
Auditor	Organisation submitted records	Read reports and audit
Communications officer	Communications tools	Templates, notices and outbox
Administrator	Workspace	Permissions and configuration

Literature and design basis

Source	Relevant idea	Project application
UN DESA and World Bank	Service access and administrative context	Citizen portal with departmental processing
Hevner and colleagues	Build and evaluate an artefact	Implemented software and repeatable evidence
NIST RBAC material	Users, roles and privileges	Action permissions with record scope
Davis and Heeks	Acceptance and deployment context	A pilot remains part of future evaluation

Hybrid delivery and evaluation

Project initiation and review gates

PID and business case, supervisor approval, SRS, architecture decisions and formal models.

Three sprints and weekly reporting

Backlog, planned burndown, supervisor minutes and reviewed work through 27 November.

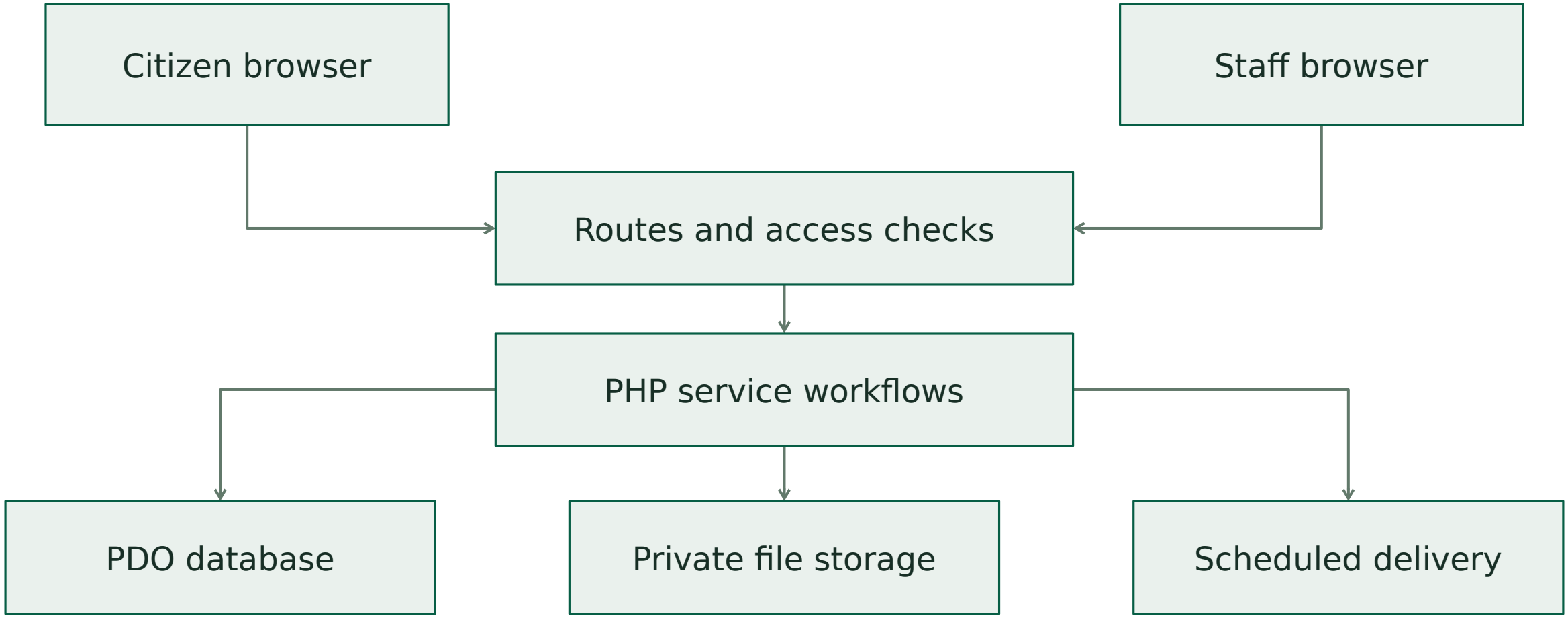
Source review and repeatable checks

Issues and pull requests connect requirements, changes, tests and individual contributions.

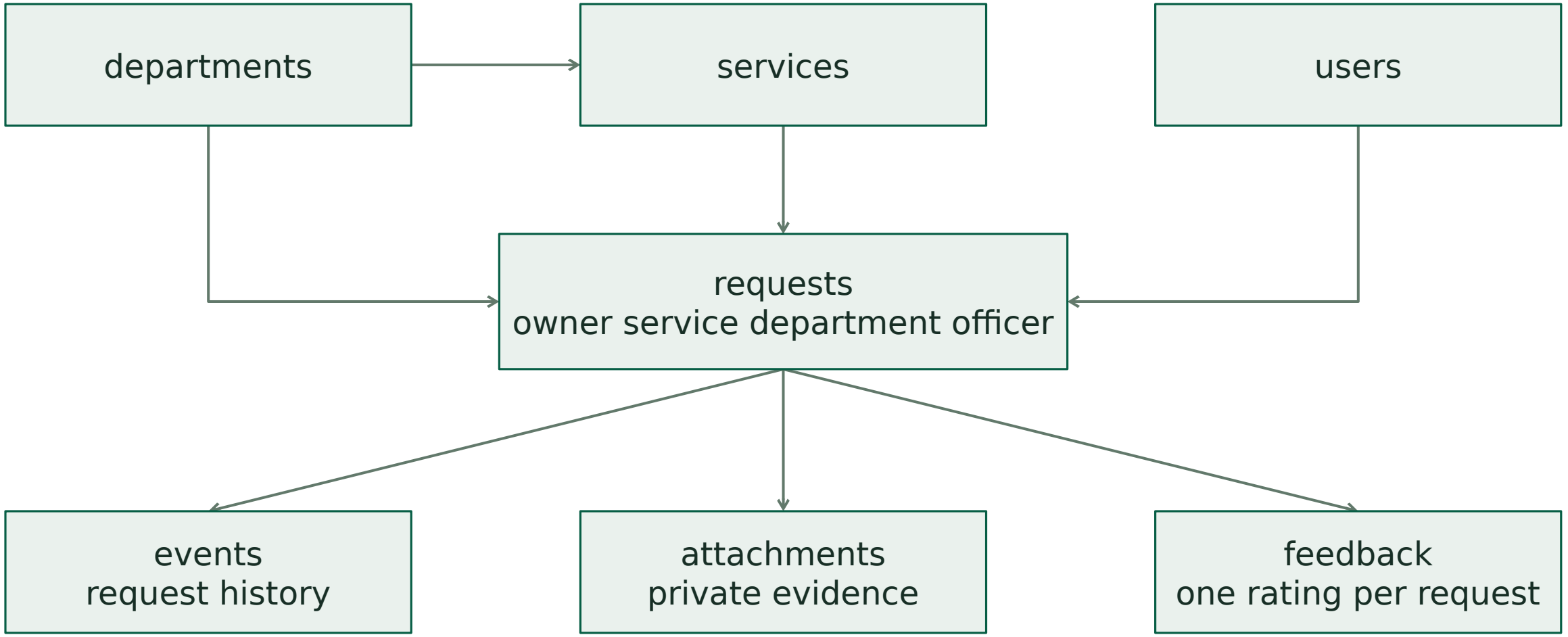
Deployment acceptance

Verify the live build independently, complete hosted checks and rehearse the 28 November defence.

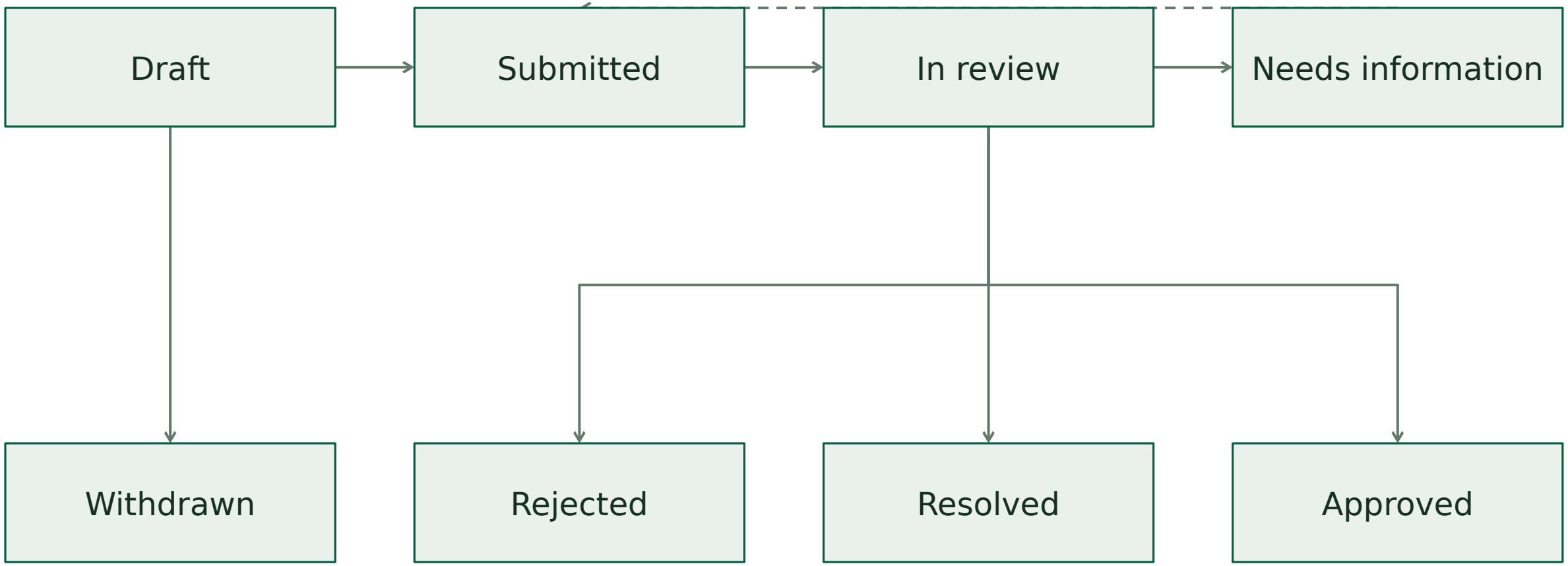
System architecture



Core data model

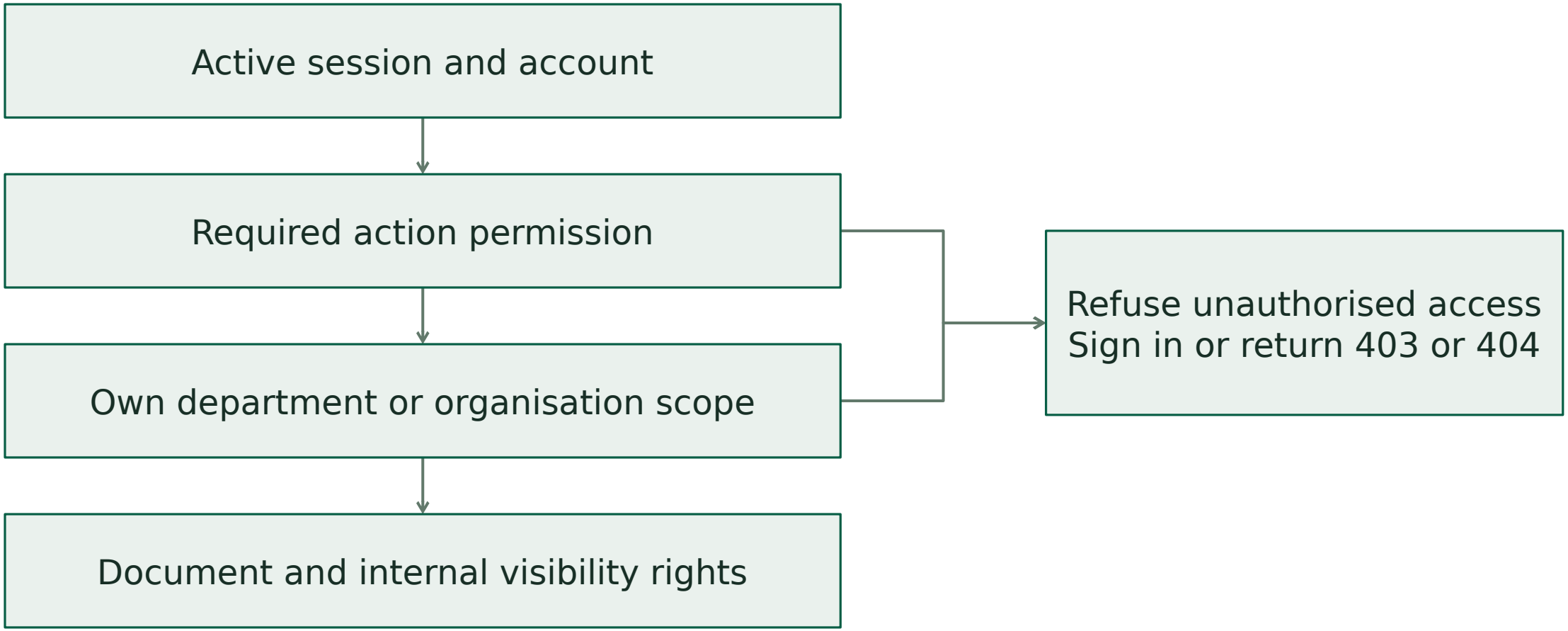


Complaint and permit lifecycle



Citizen replies return to Submitted. Resolved complaints may reopen with a fresh target.

Permission and record access checks



Account and document protection

Account protection

Password hashing, single-use tokens, authenticator MFA and device revocation.

Private evidence

MIME and size checks, random stored names, scope and internal-note visibility.

Controlled updates

CSRF validation, prepared database calls, transactions and version checks.

Output safety

Escaped citizen text, sanitised email HTML and protected CSV values.

Notifications and citizen assistance

Durable communication record

In-app notices and an outbox retain the update and delivery state.

Failure handling

Disabled channels stay held. Failed attempts use backoff and stop after five.

Operator configured connections

SMTP email and Termii SMS require the organisation’s own accounts.

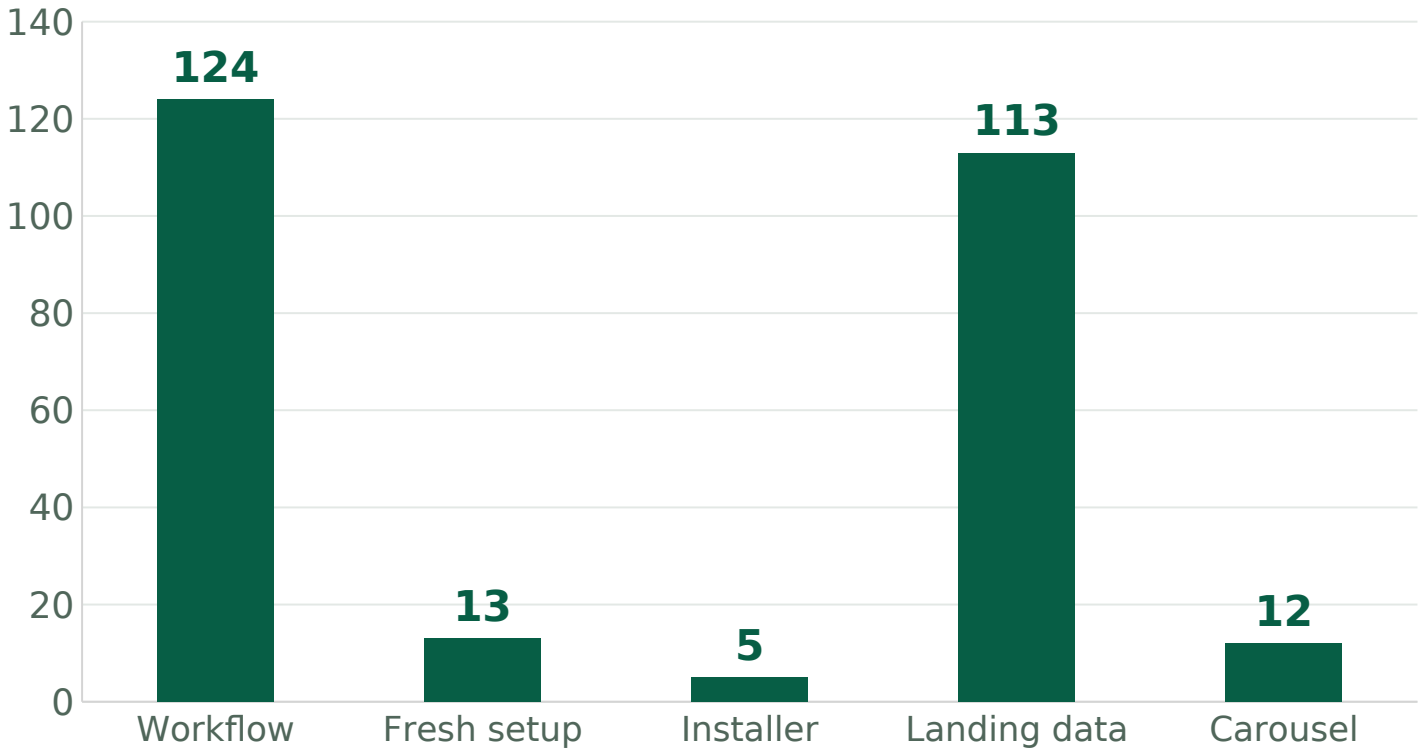
Service guidance

Local FAQs remain available. Optional AI has no private-case or decision tools.

Recorded test results

267

recorded automated assertions passed



Recorded local assertions. Carousel checks use a fixture. Hosted acceptance remains a separate gate.

Release 1 0 4 and quality evidence

Installer and packaging corrections

Protected runtime folders, the 72 byte password boundary and actionable setup diagnostics.

Public discovery and evaluation records

Three image landing page and additive preparation of 26 accounts and 36 cases across roles.

Current documentation and evidence

Report, SRS, PID, ADD, manuals and repository describe application 1.0.4 and its recorded checks.

Deployment and operational setup

New host location and database

PHP extensions, valid TLS, private writable storage and a fresh schema.

Locked installation

Generate a private setup key locally, use the exact base URL and choose your administrator.

Configured service operations

Review departments, permissions, requirements, fees, targets and notices.

Hosted acceptance and maintenance

Verify providers, cron and private routes, then prove backup restoration.

Demonstration sequence

Citizen submission and officer review

Create a complaint, request details and return a citizen reply.

Permit and private evidence

Upload a harmless PDF, record approval and show public verification.

Direct access denial

Try another citizen and unrelated department, including an attachment URL.

Communication and oversight

Show accurate outbox state, local guidance, reports and test evidence.

Limitations and further evaluation

Independent live acceptance

Record hosted MySQL, HTTPS, private routes, cron and authorised evaluation access.

Outstanding quality evaluation

Measure workload and restoration, review security and obtain consent for user evaluation.

Possible extensions

Authorised identity, payment reconciliation, signed records and retention automation.

Conclusion

CivicFlow implements a traceable route from submission to a recorded departmental outcome.

The release includes the application, PID, SRS, ADD, report, manuals and reviewable technical evidence.

Independent live access, reviewed contributions and completed project records support final acceptance.

Questions

Selected references

United Nations DESA 2024 UN E Government Survey

World Bank GovTech Maturity Index

Hevner and colleagues 2004 Design science in information systems

Davis 1989 Perceived usefulness and perceived ease of use

NIST Role based access control

OWASP 2025 ASVS 5.0.0 and application verification

NIST 2022 SSDF 1.1 and W3C 2024 WCAG 2.2

Complete bibliography and implementation evidence are included in the academic report.